

QKD: Core Concept



- **Goal:** Let two parties (Alice and Bob) share a secure secret key over a public channel.
- **Security:** Guaranteed by quantum mechanics—not just technical constraints.
- **Implication:** Security holds unless quantum mechanics is disproven.
- **QKD Protocol Phases:**
 - **Quantum Transmission Phase:** Exchange or measure quantum states (qubits).
 - **Classical Post-Processing Phase:** Convert noisy raw data into a secure key using classical algorithms.

Types of QKD Protocols



1. Prepare-and-Measure (e.g., BB84)

- Alice prepares quantum states.
 - Sends them to Bob over a quantum channel.
 - Bob measures them.
- **Key Insight:** Every prepare-and-measure protocol has an equivalent entanglement-based version.
 - Helps with proving security!

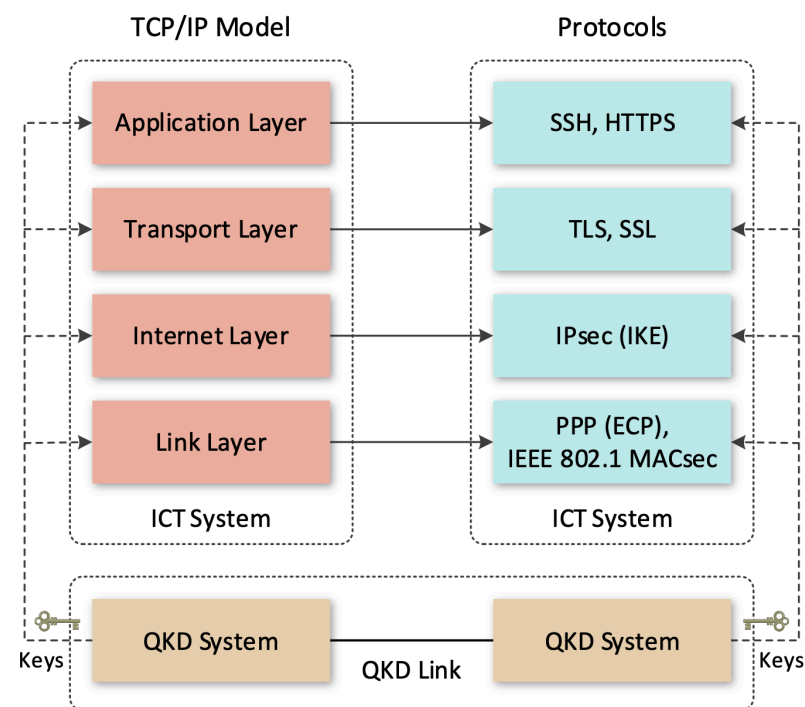
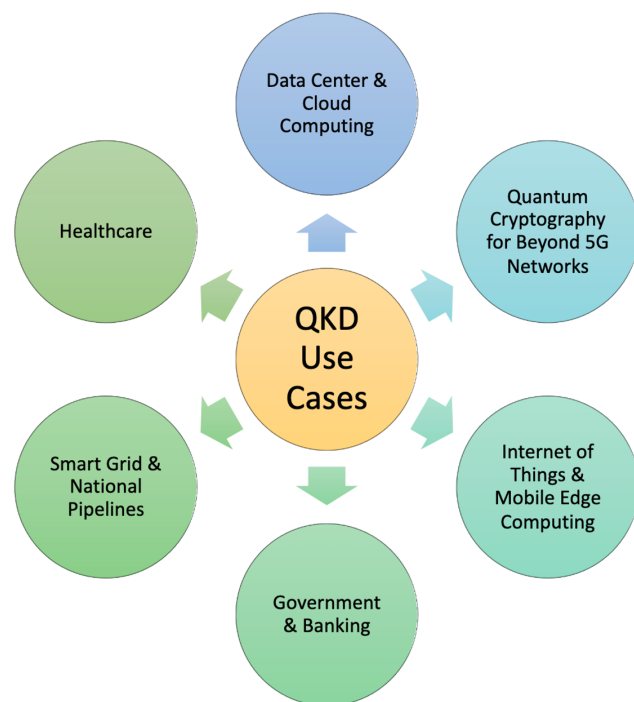
2. Entanglement-Based Protocols

- Alice and Bob share entangled qubit pairs.
- Each measures their part locally.
- No direct quantum channel needed between them.

QKD Applications

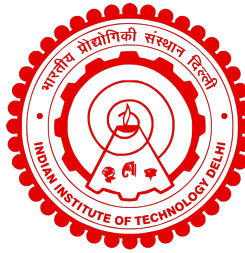


- QKD offers unconditional **information theoretic security**
- Various use cases in future networks: **Commercial and Govt.**
- Uses: Symmetric key encryption or **one-time pad** based PHY security



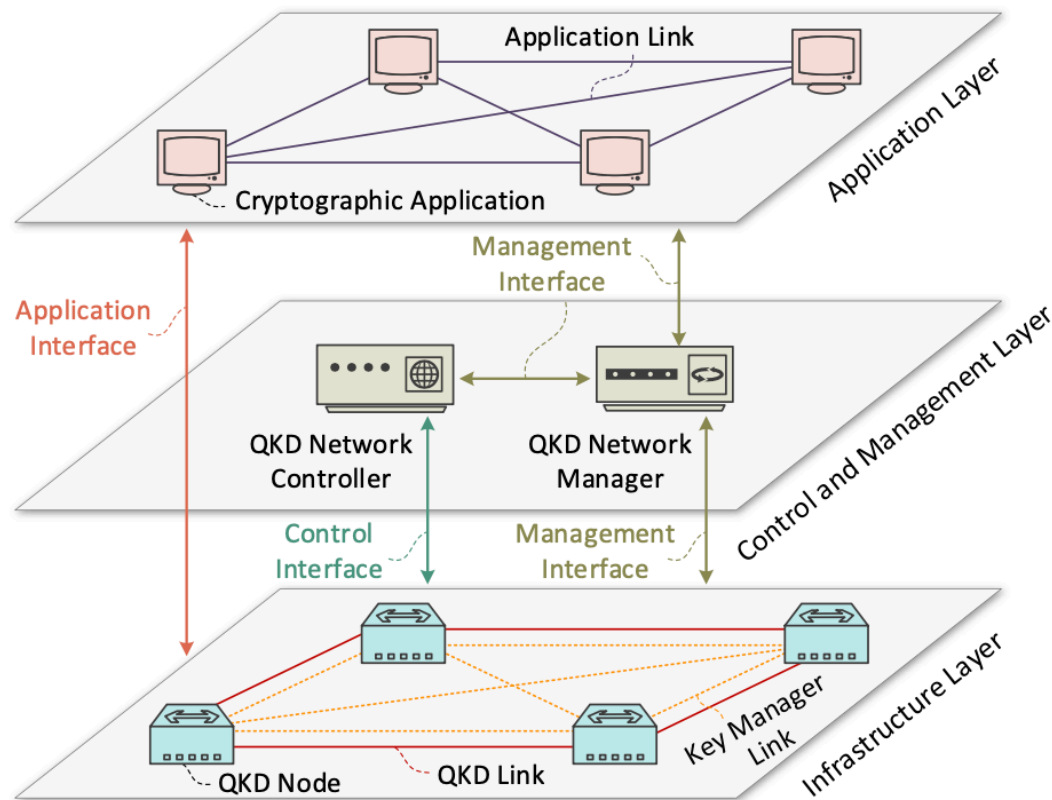
R. Liu, G. G. Rozenman, N. K. Kundu, D. Chandra, D. De , "Towards the industrialisation of quantum key distribution in communication networks: A short survey," *IET Quant. Comm*, June, 2022.

Y. Cao, Y. Zhao, Q. Wang, J. Zhang, S. X. Ng and L. Hanzo, "The Evolution of Quantum Key Distribution Networks: On the Road to the Qinternet," in *IEEE Communications Surveys & Tutorials*, vol. 24, no. 2, pp. 839-894, Secondquarter 2022, doi: 10.1109/COMST.2022.3144219.

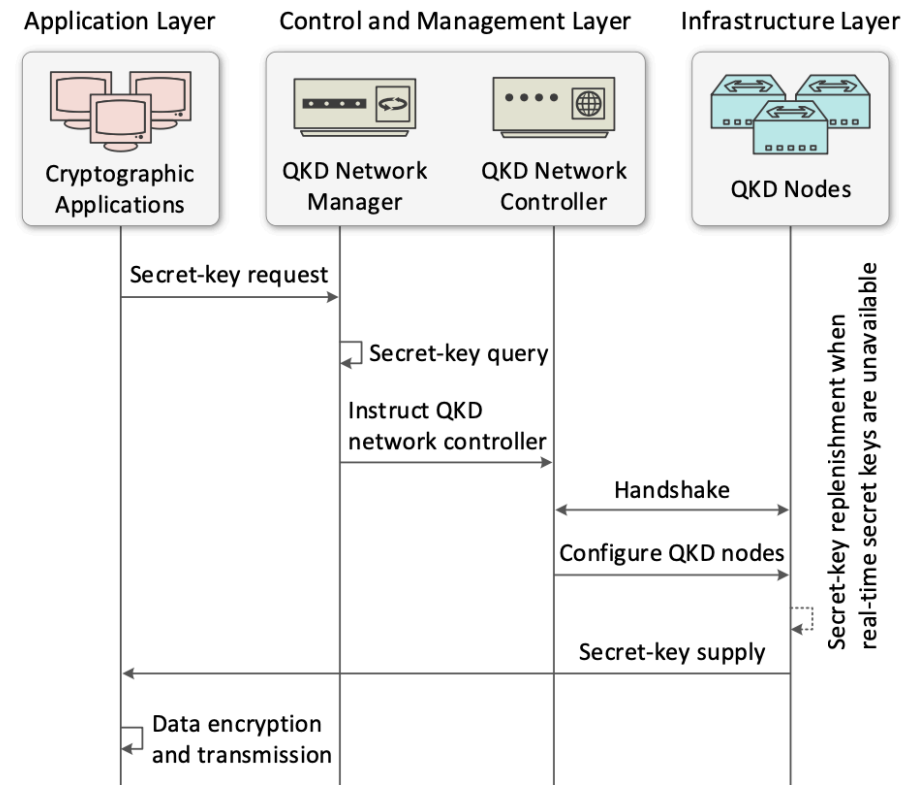


QKD Network

- Layered QKD Network architecture similar to classical networks



General architecture of QKD networks.



Workflow of service provision for cryptographic applications.

Prepare-and-Measure: Basic Setup (BB84)



- **What is it?**
 - Alice prepares quantum states (e.g., photon polarizations).
 - She sends them to Bob using a **quantum channel**.
 - Bob measures them to obtain raw key bits.
- **Channels Involved:**
- **Quantum Channel (Q):**
 - Used to send quantum states.
 - Eve can interact freely with the quantum transmission (limited only by physics).
- **Classical Channel (C):**
 - Authenticated channel used to send classical messages.
 - Eve can eavesdrop but cannot alter the messages.

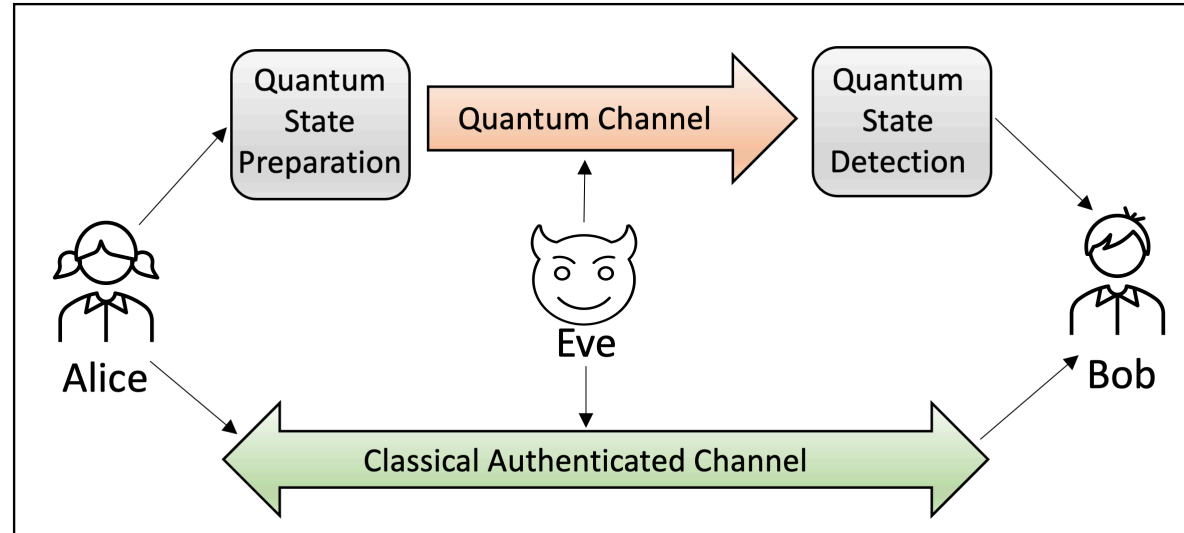
Security Assumptions and Schematic



- **Security Assumptions:**

- The classical channel is **authenticated**, ensuring Alice and Bob know whom they are talking to.
- Even if Eve listens, she cannot impersonate or modify communication.
- Without authentication, Eve could replace Bob and compromise the key.

- **Prepare-and-Measure Setup:**





BB84 Protocol: Qubit Preparation

• Alice's Preparation:

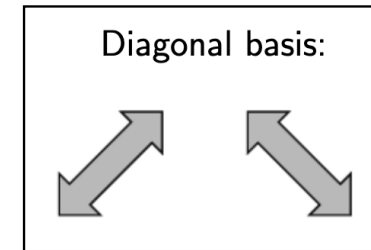
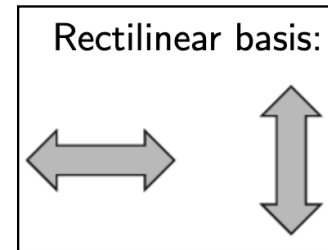
- Chooses random bit string a and basis string b , both of length $4n$.
- a_i : key bit (0 or 1), b_i : basis bit (0 = C , 1 = H)

- Prepares $4n$ qubits: $|\psi_A\rangle = \bigotimes_{i=1}^{4n} |\psi_{a_i b_i}\rangle$

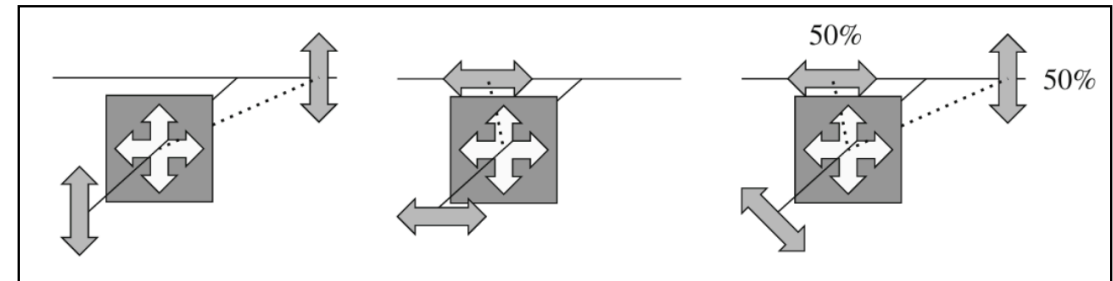
• Possible states:

- $|\psi_{00}\rangle = |0\rangle, \quad |\psi_{10}\rangle = |1\rangle$
- $|\psi_{01}\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
- $|\psi_{11}\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

• Polarisation of Photons:



• Polarization filter



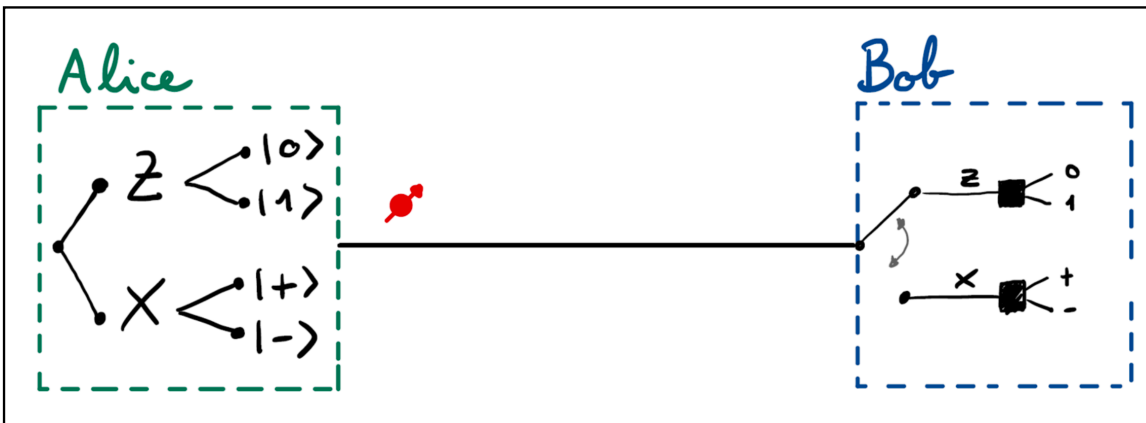
Bob Receives and Measures the State

- **Transmission:**
 - Alice sends $|\psi_A\rangle$ over the quantum channel to Bob.
 - Eve may interact with qubits but cannot know the basis b .
- **Bob receives the quantum state:**

$$\mathcal{E}(|\psi\rangle_A \langle\psi|), \quad \text{where} \quad |\psi\rangle_A = \bigotimes_{i=1}^{4n} |\psi_{a_i b_i}\rangle_A$$

- **Bob measures the state to get bit strings:**

$$a' = (a'_1, a'_2, \dots, a'_{4n}) \quad b' = (b'_1, b'_2, \dots, b'_{4n})$$



Alice

		Z		X		
		$ 0\rangle$	$ 1\rangle$	$ +\rangle$	$ -\rangle$	
Bob	Z	$ 0\rangle$	1	0	$\frac{1}{2}$	$\frac{1}{2}$
		$ 1\rangle$	0	1	$\frac{1}{2}$	$\frac{1}{2}$
	X	$ +\rangle$	$\frac{1}{2}$	$\frac{1}{2}$	1	0
		$ -\rangle$	$\frac{1}{2}$	$\frac{1}{2}$	0	1